

Integrating GenAI, LangChain, Advanced RAG & Enterprise Patterns – Level 2 Course Outline

1. Duration

9 Hours (3 x 3 Hour Sessions)

2. Objectives

Understand advanced evaluation techniques for measuring RAG system quality.
Apply optimization strategies to improve retrieval relevance, response quality, latency, and cost.
Implement security, governance, and responsible AI controls for enterprise RAG systems.
Understand operationalization of RAG solutions using LLMOps practices.
Explore modern RAG architectures and advanced retrieval patterns.
Design multi-modal and enterprise knowledge assistant solutions.
Apply best practices for scaling RAG systems from PoC to production.

3. Audience

Developers, Data Engineers, AI Engineers, Architects, QA Engineers, Tech Leads, Product Managers, Innovation Teams, Enterprise Solution Teams

4. Pre-requisite

Completion of Level 1 workshop or equivalent knowledge in GenAI & LLM Fundamentals, LangChain basics, RAG pipelines, Vector Databases, Embeddings and Similarity Search. Basic Python knowledge preferred.

5. Hardware & Network Requirements

Desktop / Laptop with minimum 16 GB RAM
Stable Internet connection (minimum 10 Mbps per user)
Local admin access

6. Software Requirements

Windows / Linux / Mac OS
Python 3.11+
VS Code / Jupyter Notebook
LangChain environment setup
Chroma / Pinecone / FAISS access
OpenAI / Azure OpenAI / Anthropic / Gemini API access
Git Client

7. Outline

Day 1

Module-1: RAG Evaluation Techniques

Why evaluation is critical for production RAG systems
Common failure modes: hallucination, poor retrieval, irrelevant chunks, incomplete answers
Retrieval metrics: Precision@K, Recall@K, MRR, MAP
Generation metrics: Faithfulness, Relevance, Groundedness, Answer correctness
Human evaluation vs automated evaluation
Golden datasets and benchmark creation
Tools: RAGAS, LangSmith, DeepEval
Lab / Demo: Evaluate an existing RAG chatbot using sample queries and identify failures.

Module-2: RAG Optimization Techniques

Chunking strategies: fixed, recursive, semantic, parent-child retrieval
Embedding model selection
Hybrid search (keyword + vector)
Re-ranking models
Prompt optimization for RAG
Cost and latency optimization
Lab / Demo: Improve retrieval relevance using chunking and reranking.

Day 2

Module-3: Security & Governance for RAG

Prompt injection risks
Sensitive data leakage scenarios
Retrieval poisoning risks
Access control and role-based retrieval
PII masking and redaction
Audit trails and observability
Responsible AI controls
Governance checkpoints for enterprise deployment
Lab / Demo: Secure enterprise document assistant design walkthrough.

Module-4: Operationalizing RAG Systems (LLMOps)

Dev to Test to Production lifecycle for RAG systems

Prompt versioning

Dataset and embedding version control

CI/CD concepts for GenAI applications

Monitoring: latency, token usage, cost, retrieval quality

Hallucination management strategies

Feedback loops and continuous improvement

Incident handling for GenAI applications

Tools: LangSmith, MLflow, Arize Phoenix, Weights & Biases

Lab / Demo: Production-grade RAG reference architecture.

Day 3

Module-5: Advanced RAG Architectures

Naive RAG vs Advanced RAG

Hybrid RAG

Graph RAG

Corrective RAG (CRAG)

Adaptive RAG

Multi-hop retrieval patterns

Agent-assisted retrieval concepts

Lab / Demo: Choose right RAG architecture for business scenarios.

Module-6: Multi-modal RAG & Enterprise Knowledge Assistants

Text + PDF + Image + Table retrieval

OCR and document intelligence pipelines

Multi-source enterprise search assistants

Use cases: HR knowledge bot, IT support assistant, Legal assistant, Sales enablement bot, Internal enterprise search

Personalization and memory concepts

Lab / Demo: Enterprise assistant solution blueprint.

Module-7: Roadmap & Best Practices

Moving from PoC to Pilot to Production

Build vs Buy decisions

Cost governance for enterprise AI

Reusable enterprise RAG patterns

Team operating model for GenAI adoption

Future trends in enterprise RAG systems

Final Discussion: Q&A; and participant solution design exercise.